Review Resources

In the first two weeks of class we will review some important concepts from Quantum Mechanics I that will be useful throughout the course.

You should feel comfortable with **Dirac notation** and **time-evolution** of quantum states. We will also briefly review the **WKB approximation**, mostly because this is a useful example of an approximation method that is widely used in quantum mechanics, but that is very different from the kinds of approximations that we will study later in the course (perturbation theory).

The following is a reading list for our review of Dirac notation and time-evolution.

- **Dirac notation and wavefunctions:**
  - Sakurai and Napolitano, Sections 1.2, 1.6, and 1.7. These must become second nature to you.
  - Additional reading: Shankar, Chapter 4 and 6 (you can read these entirely).
- **Time-evolution:**
  - Sakurai and Napolitano, Sections 2.1 and 2.4.
  - Schrödinger and Heisenberg pictures: Sakurai and Napolitano, Section 2.2.
  - See also the propagator approach in Sections 2.6 and the density matrix approach in Section 3.4.
  - Additional reading: Shankar, Section 4.3.
- **Standard Problems:**
  - Sakurai and Napolitano, parts of Section 2.5 but mostly the elementary problems in this chapter (free particle case, finite and infinite potential wells, and the harmonic oscillator).
  - Additional reading: Shankar, Chapter 5. If you understand "Section 5.4 - A Problem in Scattering" you are in good shape (we will do this in class and later revisit it with more machinery).